MIL–P–46425A(MU)
18 FEBRUARY 1964
SUPERSEDING
MIL–P–46425(ORD)
9 JANUARY 1961

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M35
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of percussion primer designated as Primer, Percussion, M35.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS
MILITARY

MIL–A–2550 —Ammunition and special Weapons; General Specification for.
MIL–I–45208—Inspection Requirements, General Specification for.
MIL–I–45607—Inspection Equipment, Supply and Maintenance of.

STANDARDS
MILITARY

MIL–STD–105D—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–109 —Quality Assurance Terms and Definitions.
MIL–STD–1235—Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS
ORDNANCE CORPS

8796522—Marking Diagram and Sealing for Wood Packing Boxes.
8840536—Primer, Percussion, M35, Assembly.

PUBLICATIONS
ORDNANCE CORPS

IEL–8840536—Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71–78—Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

FSC 1375

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:55Z
Check the source to verify that this is the current version before use.

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71–78 (latest revisions)".)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Primer. The primer shall comply with all requirements specified on Drawing (dwg.) 8840536 and with all requirements specified in applicable specifications.

3.3 Functioning (see 6.4). The primer shall function under an applied energy of 31.36 inch ounces, maximum, without evidence of squibbing or hang fire. The primer shall not function under an applied energy of 5.76 inch ounces, minimum. Testing shall be as specified in 4.3.2.

3.4 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208. The contractor's quality assurance system shall contain a procedure for the reporting of critical defectives found during the course of his inspection and during inspection by the Government. Any critical defect encountered, unless annotated by an asterisk (*) to the right of the characteristic code number (see 6.2), will be reported within 48 hours. A monthly summary report covering those critical defects annotated by an asterisk will be furnished. The agencies to whom the reports will be furnished are identified in paragraph 6.8. The reports shall contain as a minimum, the following information:

(a) Contractor and contract number.
(b) Number and revision date of specification.
(c) Characteristic code number.
(d) Lot or serial number.
(e) Number of defectives found.
(f) Assignable cause or analysis of concurrence.
(g) Corrective action taken or recommended.

When the corrective action is beyond the control of the contractor or the scope of the contract, the report shall contain the contractor's recommendation to the Government.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:55Z
Check the source to verify that this is the current version before use.

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
(b) Number of units of product inspected.
(c) Results obtained for all inspection performed.
(d) Drawing, specification number and date, together with an identification and date of changes.
(e) Certificates of analysis on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
(f) Number of items in the lot.
(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of primers produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance or fit, as determined by the Government, shall not necessitate changing the lot interfix number. Lots shall be submitted for inspection in accordance with Standard MIL–STD–105 (or Standard MIL–STD–1235, when applicable). Each lot shall contain:

(a) Parts bearing the same lot interfix number from one manufacturer.
(b) Primer mixture from one batch or blending of batches which results in homogeneity of all primer mixture used in loading the entire percussion element lot.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 *Primer, prior to assembly of cover* (see dwg. 8840536).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical: None defined.			
Major:			

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:55Z
Check the source to verify that this is the current version before use.

Categories	Defects	Method of inspection	Code No.
	AQL 0.25 percent		
101.	Charges, or portion of charge missing	Visual	01001
Minor:	AQL 0.40 percent		
201.	Evidence of workmanship (see 3.4)	Visual	01002

4.2.2.2 *Primer, prior to assembly of anvil* (see dwg. 8840536).

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:	AQL 0.25 percent		
101.	Double or multiple cover	Visual	02001
102.	Cover missing	Visual	02002
Minor:	AQL 1.00 percent		
201.	Cover damaged or improperly inserted	Visual	02003
202.	Evidence of poor workmanship (see 3.4)	Visual	02004

4.2.2.3 *Assembly* (see dwg. 8840536).

Categories	Defects	Method of inspection	Code No.
Critical:			
1.	Cup cracked or split	Visual	03001*
Major:	AQL 0.65 percent		
101.	Total length	Gage	03002
102.	Outside diameter	Gage	03003
103.	Anvil inverted	Visual	03004
104.	Anvil missing cocked, loose, damaged or malformed	Visual	03005
105.	Belled cup	Visual	03006
106.	Double anvil	Visual	03007
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	03008

4.2.3 Testing

4.2.3.1 *Composition of 70 primer mixture (when applicable)* (see dwg. 8840536)—*Major defect—Code No. 04001.* The composition of the primer mixture shall be determined for each batch or blending of batches. Three samples in sufficient quantity to perform the test shall be selected from each batch or blend of batches and shall be analyzed as specified in 4.3.1. Should the averaged values obtained for the three samples fail to meet one or more of the composition requirements specified on the drawing, the batch or blend of batches shall be rejected.

4.2.3.2 *Functioning* (see 3.3).

4.2.3.2.1 *Sensitivity—Major defect—Code No. 04002.* A sample of 700 percussion primers shall be selected from each lot for this test. If the sum of the average critical height (H average) and 5 times the standard deviation (s) multiplied by the weight of the ball exceeds the maximum requirement, or if the sum of the average critical height (H average) less 2 times the standard deviation (s) multiplied by the weight of the ball is less than the minimum requirement, the lot shall be rejected (see 6.4). The test shall be performed as specified in 4.3.2, using test equipment specified in 4.2.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:55Z
Check the source to verify that this is the current version before use.

4.2.3.2.2 *Duds—Major defect—Code No. 04003.* A sample of 300 percussion primers shall be selected from each lot for this test. Failure to function of two or more primers during this test shall be cause for rejection of the entire lot. The test shall be performed as specified in 4.3.2.2 using test equipment specified in 4.2.4.

4.2.4 *Inspection Equipment.* Index of Inspection Equipment Number IEL–8840536 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 **Test methods and procedures.**

4.3.1 *Composition of primer mixture.* The composition of the primer mixture shall be determined by performing the following tests in the sequence specified.

4.3.1.1 *Preparation of specimen.* In order to calculate the percentage of each ingredient on a moisture-free basis, the moisture content of the specimen shall be determined as follows: Approximately one gram (gm.) of the specimen shall be transferred to a tared, sintered glass, or filtering crucible having a thick asbestos mat. Using a wooden spatula, the sample shall be spread quickly in a thin layer over the bottom of the crucible, and weighed rapidly. The crucible shall be allowed to remain in a desiccator containing calcium chloride, for at least 16 hours and then weighed. The loss in weight shall be calculated to percent moisture.

4.3.1.2 *Trinitrotoluene.* To the dried sample in the crucible, 15 milliliters (ml.) of anhydrous chloroform shall be added. The solvent shall be allowed to stand in contact with the sample for 5 minutes. Then, with the end of a wooden rod, gently and carefully the lumps shall be crushed against the asbestos mat. The solvent shall be allowed to drain through the mat without suction. This extraction shall be repeated with two more 15 ml. portions of chloroform. Completeness of extraction may be tested for at this point, by adding 2 drops of the chloroform filtrate, to 2 drops of alcoholic potassium hydroxide solution (1 molar) in a white spot plate. A red coloration indicates incomplete extraction. If incomplete, extraction shall be continued with 15 ml. portions of chloroform until extraction is complete, as shown by test. Air shall be drawn through the crucible until no odor of chloroform remains. The crucible and residue shall be dried in an oven, maintained at 100 plus or minus 5 degrees Centigrade (degrees C.) for 15 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated to percent trinitrotoluene on a moisture-free basis.

4.3.1.3 *Antimony sulfide.* A 12.5 percent (weight: volume) ammonium acetate solution shall be prepared. The residue in the crucible from the trinitrotoluene determination shall be treated with 20 ml. of warm 12½ percent saturated ammonium acetate solution. The solution shall be allowed to re-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:55Z
Check the source to verify that this is the current version before use.

main in contact with the residue for 10 minutes, breaking up any lumps with the flattened end of a glass rod. By means of suction, the solvent shall be drawn off into a clean flask, and the extraction repeated with two more, 20 ml. portions of the warm solution. At the end of the third treatment, completeness of extraction may be tested for by treating 1 or 2 drops of the ammonium acetate filtrate with 2 drops of diluted nitric acid and 2 drops of slightly acidified ferric alum solution on a white spot plate. If extraction is not complete, as shown by a red coloration, treatment of the residue shall be continued with warm 10 ml. portion of the solvent. When extraction is complete, as shown by absence of red coloration when two drops of the filtrate are tested, the crucible and residue shall be washed thoroughly with water, collecting the washings with the ammonium acetate filtrate. The residue shall be washed with alcohol and ether, and aspirated until the odor of ether is no longer detectable. The crucible and contents shall be placed in an oven, maintained at 100 plus or minus 5 degrees C., for 15 minutes, then cooled in a desiccator and weighed. From this weight, the weight of the empty crucible shall be subtracted and the difference shall be calculated to percent antimony sulfide on a moisture-free basis.

4.3.1.4 *Lead thiocyanate.* The filtrate and washings from the antimony sulfide determination shall be transferred to a 400 ml. beaker. The volume should be between 150 and 200 ml. A 2 ml. portion of glacial acetic acid shall be added and the solution heated to boiling. While stirring vigorously, 8 ml. of 5 percent potassium dichromate solution shall be added slowly to the beaker. The solution shall be stirred for 1 or 2 minutes longer, then digested on a steam bath for 1 hour, with frequent stirring. The precipitate should have a reddish-orange color. The precipitate shall be filtered through a tared filtering crucible, washed with hot water, and then the crucible and precipitate shall be dried in an oven, maintained at 100 plus or minus 5 degrees C., cooled in a desiccator and weighed. The increase in weight shall be calculated to percentage of lead thiocyanate on a moisture-free basis as follows:

Lead thiocyanate=100.05×lead chromate

4.3.1.5 *Potassium chlorate.* The sum of the percentages of trinitrotoluene, antimony sulfide, and lead thiocyanate shall be subtracted from 100 and the remainder shall be reported as percentage of potassium chlorate.

4.3.2 *Functioning.* Prior to performing these tests the primers shall be conditioned at a temperature of 75 degrees Fahrenheit (F.) plus or minus 5 degrees F., for a period of 24 hours, minimum.

4.3.2.1 *Sensitivity.* The sample of 700 primers shall be divided into 14 groups of 50 primers each and tested in the following manner:

The first group (50) of primers shall be tested by dropping a steel ball $^{15}/_{16}$ inch in diameter and weighing 1.94 plus or minus .02 ounces on the firing pin of the test fixture from a height of 9 inches. (The height of drop shall be measured as the height from the top of the firing pin to the bottom of the ball). The number of primers that fire, misfire, hangfire or squib (see 6.6) shall be recorded. This procedure shall be repeated at a successively increasing heights of 1 inch using a new group of 50 primers at each different height, until a height is reached at which all primers in the group being tested fire. The test fixture shall then be set at a height of 8 inches and the above procedure repeated at successively decreasing heights of 1 inch using a new group of 50 primers at each height, until a height is reached at which all primers in the group being tested fail to fire. Acceptance or rejection of the lot shall be determined from calculations based on recorded data (see 6.4).

Note: All samples subjected to test shall be scrapped. Any remaining untested samples may be returned to the lot from which they were taken.

4.3.2.2 *Duds.* Each primer of the sample of 300 primers shall be tested by dropping a stell ball $^{15}/_{16}$ inch in diameter and weighing

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:55Z
Check the source to verify that this is the current version before use.

1.94 plus or minus .02 ounces on the firing pin of the test fixture from a height of "16 inches".

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Level C.* Percussion primers shall be packaged in accordance with the requirements contained in Code of Federal Regulations 49 CFR 71–78.

5.2 Packing

5.2.1 *Level C.* Primers shall be packed in accordance with Code of Federal Regulations 49 CFR 71–78.

5.3 Marking. Unit containers shall be printed, stamped or labeled as follows:

(QTY PRIMER, PERCUSSION M35
LOT NO.
DATE PACKED (MO/YR)

Letters shall be all capitals, minimum ¼ inch in size. Ink shall be commercial black, waterproof and label paper shall be white paper having a smooth finish. Exterior containers shall be marked in accordance with Drawing 8796522. The I.C.C. nomenclature shall be "PERCUSSION PRIMER–HANDLE CAREFULLY".

5.4 Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 *Inspection Equipment Lists.* Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6. for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection Equipment List codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF	—Contractor's design responsibility on Ordnance format; ORDM–608–12.
CDCF	—Contractor's design responsibility on contractor format.
OD	—Ordnance design.
ODMU	—Ordnance design, mandatory for use.
CE	—Commercial equipment.
SCD	—Specification control Drawing.

6.3.3 *Ordnance designs.* Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and non-mandatory. When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:55Z
Check the source to verify that this is the current version before use.

alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawings.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608–12, and in addition, contain the detail and information specified above.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Formula and Method of determining H average and s.

6.4.1 Formula:

H average = A plus B plus C
s = I times the square root of L minus M

where:

A = the sum of column 3.
B = the first drop height at which all elements misfired
C = one-half the drop height increment (see 6.7).
L = the sum of Column 5.
M = the square of the sum of Column 3.
I = the drop height increment.

6.4.2 *Tabulation of data.* Divide a sheet of paper into columns and tabulate the data as follows:

(a) Title Column 1 "Height of Drop" and list all drop heights in consecutive order starting with the lowest height. DO NOT list the heights at which all elements fired or misfired.

(b) Title Column 2 "Number of Failures" and list the sum of the num-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:55Z
Check the source to verify that this is the current version before use.

ber of elements that misfired, squibbed or hang fired (see 6.6) at each intermediate height.

(c) Title Column 3 "Decimal Fraction of Elements Failing to Fire at Each Intermediate Height" and record the decimal fraction that failed to fire at each intermediate height. This number is obtained by dividing the value obtained under "Number of Failures" by the number of elements tested at that height. Calculations should be carried out and recorded to the nearest one-hundredth (.01).

(d) Title Column 4 "Standard Deviation Factor" and list odd numerals in sequence starting with 1, for as many drop heights as are listed in Column 1.

(e) Title Column 5 "Standard Deviation Factor Times Fraction of Failures" and, by multiplying each decimal fraction in Column 3 by the number on the same line in Column 4, enter the product on the corresponding line in Column 5.

6.4.3 Procedure for solving equations:

(a) To obtain "A"—Add the values in Column 3 and substitute the sum for "A".

(b) To obtain "B"—The first drop height at which all elements failed to fire and substitute the product for "B".

(c) To obtain "C"—Divide the drop height increment by 2 and substitute the dividend for "C".

(d) To obtain "L"—Add the values in Column 5 and substitute the sum for "L".

(e) To obtain "M"—Square the sum of Column 3 and substitute the sum for "M".

(f) To obtain "H average"—Add the values substituted for A, B, and C.

(g) To obtain "s"—Subtract the value substituted for "M" from the value substituted for "L" and then obtain the square root of the difference and then multiply the square root by the height of increment.

(h) Using the calculated values for "H average" and "s", if "H average" plus 5 "s" multiplied by the maximum weight of the ball is equal to or less than the maximum inch ounce energy value required, and "H average" minus 2 "s" multiplied by the minimum weight of the ball is equal to or greater than the minimum energy value required, the lot should be considered acceptable for sensitivity.

6.5 Inch ounces. Inch ounces are defined as the drop height in inches multiplied by the weight of the ball in ounces.

6.6 Failure. Defined as any one of the following primer malfunctions:

(a) Misfire—a failure wherein initiation of the primer mixture does not occur.

(b) Squib—a failure wherein a burning of the primer mixture occurs without detonation.

(c) Hangfire—a failure wherein an audible delay occurs between the instant of application of energy and the detonation report.

6.7 Check test for possible deterioration. The following test is designed to prevent the loading of deteriorated primers into major ammunition items. The test should be performed on primers from each lot which has been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the primers are loaded does not require that the check test be performed, the contracting officer should insert provisions for the performance of the check tests in the contract. The contracting officer should arrange to have the contractor reimbursed for any expenses incurred in the performance of the check test for possible deterioration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:55Z
Check the source to verify that this is the current version before use.

6.7.1 *Sensitivity.* The test shall be performed as specified in 4.3.2.1, using test equipment specified in 4.2.4. A sample of 1050 primers shall be selected from each lot for this test. Seventy five primers shall be tested at each height. If H average plus 4.66 s exceeds 16 inches or if H average minus 1.86 s is less than 3 inches, the lot shall be rejected.

6.7.2 *Duds.* The test shall be performed as specified in 4.3.2.2. A sample of 300 primers shall be selected from each lot for the test. Failure to function of 3 or more primers shall reject the lot.

Custodian:
Army—MU

6.8 Submission of critical defect reports. For U.S. Army Munitions Command procurements, one copy of each critical defect report will be furnished the cognizant Government inspection agency; the Army Ammunition Procurement and Supply Agency, ATTN: SMUAP–RO; Picatinny Arsenal, ATTN: SMUPA–ND.

6.8.1 *Reporting to other services.* For other than U.S. Army Munitions Command procurements, critical defect reporting shall be as prescribed by the Contracting Officer.

Preparing activity:
Army—MU

Project No. 1375–A686

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:55Z
Check the source to verify that this is the current version before use.